

"Imitating at the Last Minute": Exploring Approaches to Enhance Public Speaking Performance in Limited Time

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Abstract

Public speaking is essential in our daily lives but often induces anxiety, particularly in unexpected situations with limited preparation time. This study proposes two video-based methods to enhance speaking performance within a limited time: a content-modified video (CmV) with a professional speaker delivering the same script of the upcoming speech, and a self-video (SV) featuring the user delivering the presentation. In the pilot user study, 14 participants give presentations before and after viewing three kinds of videos: original video (OV), CmV and SV. We evaluate speech performances, public speaking anxiety, acceptance and perception of these methods using questionnaires and interviews. The results indicated that viewing SV increased public speaking confidence significantly more than viewing OV. In the CmV condition, participants' speech pace improved significantly more than in the OV condition. We considered that CmV and SV were beneficial for individuals, with participants' chosen approach dictated by their strategies.

CCS Concepts

• Human-centered computing • Human computer interaction(HCI) • Empirical studies in HCI;

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1 INTRODUCTION

Speech is an important means of expressing thoughts, feelings, and ideas. It is essential in various contexts, such as classroom presentations, corporate reports, and storytelling, to convey information and emotions. Nevertheless, Public speaking often causes anxiety, particularly when faced with unexpected presentations or limited preparation time. In fact, fear of public speaking is one of the most common social anxieties, with symptoms like nervousness, rapid heartbeat, and stuttering[7, 23]. This anxiety arises from several factors, such as poor preparation, fear of failure, lack of suitable role models, and insufficient speaking skills [1, 2, 27].

Due to this fact, there are many strategies to reduce anxiety, increase self-confidence and improve public speaking skills. For example, thorough preparation, including topic understanding and practice, can boost confidence [16]. As mentioned in [5, 22], viewing self-modeling videos of presentations can also reduce nervousness. Gradual exposure and desensitization therapies can also alleviate anxiety and improve public speaking performance [17]. In addition to these strategies, researchers also applied a variety of techniques to public speaking. Many researchers utilized Virtual Reality (VR) and Augmented Reality (AR) technology to reduce public speaking anxiety by facilitating exposure therapy[4, 6, 13, 14]. As shown

in [10, 20, 28, 30], virtual agents also provided speakers with support to improve public speaking skills and reduce their anxiety levels. Additionally, several studies offered platforms or systems that provided feedback and suggestions for public speaking, thereby assisting users in improving their speaking abilities [12, 31, 33].

Many studies on public speaking focused on reducing anxiety and enhancing skills through a period of training. However, in reality, individuals often encounter unexpected speaking tasks that require the enhancement of their speaking performance within limited time. In order to address this situation, this study proposed two video-based practice methods to enhance speaking performance within limited time. The first method was to generate a video that contained the same script as the upcoming speaking using voice cloning, which assisted users in directly imitating speaking skills in the video. The second method was to generate a self-video using voice cloning and face-swapping, which made users feel as if they had already completed the presentation. In addition, we offered a control video: an original speaking video from the Internet. 14 participants were invited to join in this experiment to rehearse their speeches by viewing the three kinds of videos. During the experiment, we recorded videos from participants before and after viewing videos and invited professional teachers to evaluate their speech performance. We also collected feedback from the participants using questionnaires and semi-structured interviews.

The core contributions of this work are summarized as follows: (1) investigating approaches to enhance the performance of public speaking within limited time; (2) early evidence from a pilot study that explores the usefulness of these approaches in public speaking preparation; (3) an overview of a general perception of public speech preparation and acceptability of these different approaches.

2 RELATED WORK

2.1 Face-swapping and voice cloning

Face-swapping uses deep learning algorithms to replace an individual's facial features with those of another, creating a realistic "face-swapping" effect. Deep learning models such as Generative Adversarial Networks (GAN) extract and blend facial characteristics, producing the final face-swapped pictures or videos by image synthesis. Face-swapping applies across various fields, including education [8] and healthcare [15, 24, 32, 34, 36]. Wang et al. [8] utilized the face-swapping technology to generate self-modeling videos to teach children with autism hand-washing skills, which enhanced the motivation of children and facilitated the generalization of skills. Moreover, in the field of healthcare, face-swapping technology served as a soft biometric deidentification method to protect privacy [15, 24, 32, 34, 36]. Li et al. [15] specifically explored the application of face-swapping for anonymizing patients in videos of epileptic seizures while retaining clinically useful seizure semiology.

Voice cloning utilizes text-to-speech TTS technology and deep learning algorithms to generate a synthetic replication of a person's voice and capture their speech patterns, intonation, and unique characteristics. The voice model can then generate speech similar to that of the original speaker. Voice cloning is utilized in different fields, such as assistive technology [9], healthcare [11, 29], and education [21]. Jin et al. [9] introduced a method for converting speech

across multiple accents, which preserved the speaker's original voice characteristics while enabling accurate accent transformation. This approach may not only enhance communication by allowing individuals to understand and interact with speakers from diverse linguistic backgrounds but also support language learning by providing learners with more authentic, varied accent exposure.

2.2 Approaches to improve public speaking performance

In order to assist individuals in their preparation and practice for public speaking, several researchers have developed applications. Wu et al. [33] introduced a system named Trinity, designed to enhance academic oral presentations. This system aided foreign language students in delivering more natural presentations, reducing monotony, and significantly increasing the attractiveness of their speeches. The VoiceCoach was proposed by Wang et al. [31] to assist public speakers in proficiently mastering the modulation of voice qualities. By evaluating high-quality voice samples from TED Talks, the application provided users with immediate feedback and personalized recommendations for improving their presentations.

In recent years, AR and VR have emerged as powerful tools for overcoming public speaking anxiety. Leong et al. [14] found that AR face filters had the potential to aid in alleviating anxiety in online public speaking. LeFebvre et al. [13] conducted a study examining the application of VR in public speaking, revealing that VR technology mitigated public speaking anxiety by merging real and virtual interactions. Van Ginke et al. [6] examined the impact of VR technology on improving students' oral presenting skills by offering comprehensive feedback.

As shown in [28], there are some studies introducing virtual agents to mitigate public speaking anxiety. Murali et al. [20] provided a virtual agent referred to as Friendly Face, which was able to perceive the behavior of the speaker accurately and offered both emotional and instrumental support. The results indicated that participants utilizing the agent exhibited markedly reduced public speaking anxiety, as evidenced by both physiological and self-reported metrics. Wang et al. [30] developed an Amazon Alexa-based virtual coach to minimize public speaking anxiety. It reduced public speaking anxiety by helping users discover and change negative thought patterns through cognitive reconstruction exercises.

Moreover, some studies investigated the utilization of wearables to provide speakers with real-time feedback to improve their performance [3, 25]. According to [25], Rhema was designed to help people improve their public speaking skills. By detecting the speaker's volume and rate of speech, Rhema provided instant feedback during the presentation, assisting speakers in adapting their presentations.

3 METHOD

Before starting the experiment, we prepared three different types of videos for participants to watch and learn from. The first type of video was an unprocessed speech video downloaded from the Internet, called original video (OV). The second type of video was a content-modified video (CmV) in which the script was identical to that of the upcoming speech, utilizing GPT-SoVITS and MuseTalk for speech generation. The third type of video was a self-video

(SV) in which the speaker was the participant, and the script was also the same as that of the upcoming speech, employing GPT-SoVITS, MuseTalk and FaceFusion for face and speech generation. The following is a concise overview of the GPT-SoVITS, MuseTalk, FaceFusion and generation methodologies of CmV and SV.

GPT-SoVITS¹ integrates Variational Autoencoders with GPT models to enable high-fidelity, few shot, and real-time voice conversion and text-to-speech generation. It utilizes self-supervised learning and adversarial training to capture voice characteristics from limited audio samples, allowing for natural and coherent speech synthesis with minimal data.

MuseTalk [35] is a real-time, high-quality audio-driven lip synchronization model that generates talking face videos by projecting the face image into a latent space encoded by a Variational Autoencoder. It leverages a multi-scale U-Net architecture to efficiently fuse audio and visual features, enabling rapid and accurate lip-sync generation for realistic speech-driven facial movements.

FaceFusion² leverages advanced models like Inswapper128FP16 and GFPGAN 1.4 for high-quality face swapping. Inswapper128FP16 ensures seamless transfer of facial features with consistent lighting and angles, while GFPGAN 1.4 restores fine details and enhances image quality, particularly in low-resolution areas, thereby improving overall face-swapping fidelity.

3.1 Generation of CmV

We collected a proper model video of public speaking from the internet. We used GPT-SoVITS and MuseTalk to generate CmV (see fig. 1).

We used an audio processing tool to extract audio from the model video, then applied GPT-SoVITS to process it in three steps. First, we pre-processed the audio by reducing noise for better quality. Next, we manually proofread the automatic speech recognition(ASR)-translated script for accuracy. In the model training phase, we extracted speaker-specific features and used GPT-SoVITS to train the voice model, adjusting parameters for optimal balance between stability and expressiveness. Finally, we generated the audio using TTS, modifying aspects like speaking rate, emotional tone, and pauses. Lastly, MuseTalk synchronized the generated audio with the mouth movements in the video.

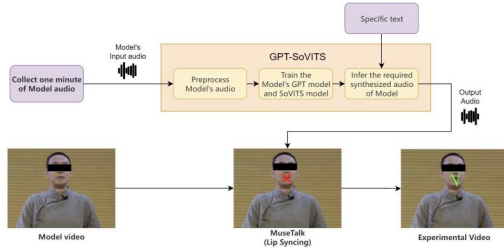


Figure 1: Generate CmV using GPT-SoVITS and MuseTalk with a model video

¹GPT-SoVITS. 2024. <https://github.com/RVC-Boss/GPT-SoVITS>.

²FaceFusion 2.6.1. 2024 <https://github.com/facefusion/facefusion>

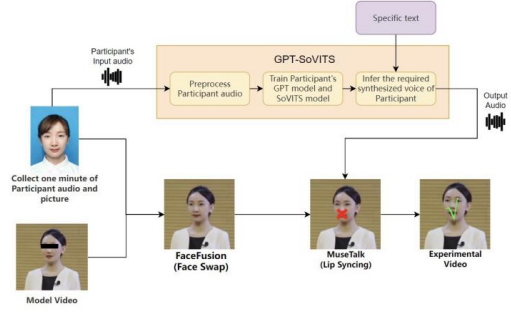


Figure 2: Generate SV using GPT-SoVITS, MuseTalk and FaceFusion Tools with a model video and participants' audio and picture

3.2 Generation of SV

We collected a one-minute speech and a photograph from each participant as materials. We used GPT-SoVITS, MuseTalk and FaceFusion to generate SV (see Fig. 2). Unlike CmV, SV generation included a face-swapping step. We replaced the speaker's face in the video with the participant's face employing FaceFusion, followed by using MuseTalk.

4 USER STUDY

4.1 Participants

We recruited fourteen volunteers (8M, 6F) in this study. All participants were native speakers of Mandarin Chinese, aged 21-40 ($mean = 27.1, SD = 5.91$). We requested them to evaluate the frequency of their public speeches using a 5-level Likert scale, with 1 indicating never giving a public speech and 5 indicating frequently giving a public speech. The result showed that the mean score was 2 ($SD = 1.11$), which means they gave public speeches with lesser frequency.

4.2 Procedure

Fourteen participants were randomly split into two groups of seven: one group with an informative topic and the other group with a narrative topic. Each group received three scripts corresponding to their respective speech type. To evaluate the complexity of the scripts, 18 college students (not involved in the experiment) rated scripts using a 5-level Likert scale, where higher scores indicated more difficulty. A one-way ANOVA analysis revealed no significant differences in the difficulty ratings between the scripts for each speech type (informative: $Mean = 1.81, SD = 0.87$; narrative: $Mean = 2.15, SD = 1.12$), which suggested that the difficulty level was consistent across all scripts within each speech category.

Three types of videos (OV, CmV and SV) were available for the participants to watch and learn from during the experiment. The average duration of these videos was 73.36s ($SD = 13.09$). Participants were given sufficient time to practice with the provided scripts. When participants were ready, they delivered their initial speech (pretest). Then they viewed one type of video and tried to learn from it. After practice, they gave a second speech (posttest). Subsequently, participants completed the questionnaire including seven rating

items about public speaking modified from the Personal Report of Communication Apprehension (PRCA-24, [18]) scale, which assesses communication apprehension in four contexts: public speaking, dyadic, group and meeting. This process was repeated for each video type, with the viewing order counterbalanced to minimize sequential and fatigue effects. After completing all speeches, the participants were asked to complete a semi-structured interview to provide feedback on these approaches. The procedure of the experiment is shown in Fig. 3. Pretest and posttest speeches were recorded, and images of the audience were placed before participants to increase immersion.

Finally, three qualified teachers were invited to evaluate the public speaking competency of participants by a questionnaire including eight items from the Public Speaking Competency Instrument (PSCI, [26]). Since the public speaking scripts were consistent among participants, we did not include the content-related items of the PSCI.

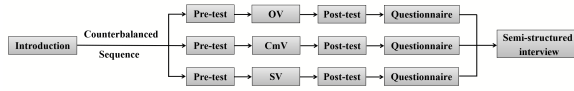


Figure 3: Experimental Procedure of the User Study

5 RESULTS

5.1 Analysis of public speaking anxiety under three conditions

As previously stated, we employed the seven items from PRCA-24 regarding public speaking utilizing a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Questions 2, 4, and 6 are reverse-scored. These were reverse-scored before data processing, with higher scores indicating greater public speaking anxiety. The specific questions in the questionnaire are presented in Appendix A.1.

Fig. 4 (a) shows the rating scores of seven questions under the three different conditions: OV, CmV and SV. A one-way ANOVA was conducted to assess whether different video types affected participants' speech anxiety. The result was significant for Question 6 (related to confidence) ($F_{(2,39)} = 0.97, p = 0.04$). Post hoc analysis revealed that the rating scores of the SV condition were higher than the OV condition ($p = 0.04$, SV condition: $Mean = 3.21, SD = 1.12$, OV condition: $Mean = 2.85, SD = 0.95$), suggesting that participants felt SV was more effective in boosting confidence for the upcoming speech than OV.

5.2 Analysis of public speaking performance evaluation

As mentioned before, we included eight questions from the PSCI questionnaire, excluding those related to content. We used a five-level Likert scale, wherein higher scores indicate more significant degree of agreement with the statement. The specific eight questions from the PSCI questionnaire are provided in Appendix A.2.

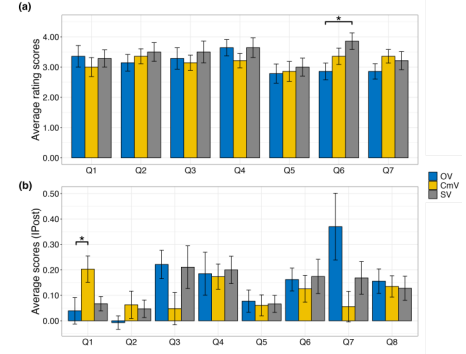


Figure 4: (a) Average Rating Scores on Questions about public speaking from PRCA-24 under Three Conditions. (b) Averaged Improvement Scores (IPost) on Selected Questions from PSCI under Three Conditions. (* $p < 0.05$)

The sequence of pretest and posttest of speech videos was randomized for all participants. The three teachers independently evaluated each video using the questionnaire. Kendall's coefficient test showed high inter-rater reliability among the three teachers for both the pretest and posttest evaluations ($ps < 0.001$, pretest: Kendall's $(w) = 0.59$, posttest: Kendall's $(w) = 0.63$). The amount of improvement from the pretest to the posttest was calculated using the following equation for the posttest score, referred to as the IPost score:

$$I_{post} = \frac{\text{posttest score} - \text{pretest score}}{\text{pretest score}}$$

Fig. 4 (b) displays the IPost scores of eight questions under the three different conditions: OV, CmV and SV. A one-way ANOVA was conducted to examine whether viewing different types of videos significantly affected public speaking performance. The results were significant for Question 1 (related to speech pace/speed) ($F_{(2,39)} = 3.75, p = 0.03$). Post hoc analysis revealed that the rating scores of the CmV condition were higher than the OV condition ($p = 0.04$, CmV condition: $Mean = 0.20, SD = 0.19$, OV condition: $Mean = 0.04, SD = 0.19$), suggesting that participants were more likely to adopt an appropriate speaking rate when learning from CmV.

5.3 Analysis of semi-structured interview

There were 3 questions in the semi-structured interview. The first question was intended to gain a general perception of which aspects are important during public speaking preparation. The second question aimed to investigate advantages and disadvantages of the CmV and SV conditions. The last question discussed the recommendations for the application of this kind of approach.

With respect to key factors in public speaking preparation, familiarity with the script and interaction with audiences, including eye contact, body language, etc., were the most stressed by participants. Other notable factors, such as speech pace, tone and rhythm were also considered as key factors. In addition, supporting materials, speaker's mindset and confidence were also mentioned by a few participants.

As for the second question, the following is a summary of the advantages of the CmV and SV: (1) since the script in the CmV was

the same as the upcoming speech, participants found the CmV provided practical guidance on speech delivery, including tone, pauses, emotion, and rhythm, allowing participants to replicate skills for better learning efficiency. (2) viewing SV helped participants to reduce public speaking anxiety and increased confidence. (3) viewing SV enabled self-reflection by allowing participants to view their performance from a third-person perspective. However, there were also some disadvantages: (1) both CmV and SV lacked variety in body language; (2) some participants noted mismatched appearance in the SV, creating an uncanny valley effect; (3) some participants noted mismatched appearance (such as hairstyle and clothing) in the SV, inducing an uncanny valley effect; (3) the voice in the SV lacked emotional depth and variation.

The following are the key suggestions proposed by participants: (1) adding a feature for customized speech scripts based on speaking styles and topics, allowing personal adjustments; (2) optimizing the speaker's body language in both videos; (3) including appearance customization options in SV condition.

6 DISCUSSION

This study investigated methods to enhance the performance of public speaking within a limited time in order to assist individuals in coping with unexpected speaking tasks. We provided three kinds of videos for participants to learn and practice: OV, CmV and SV. Professional teachers were also invited to assess the speech performance of the participants before and after practicing. From the results, we found that compared with OV, watching SV markedly increased public speaking confidence. Participants' speech pace in public speaking significantly improved in CmV condition. We also collected the demands for aided public speaking preparation and the acceptability of these methods in this paper.

Compared with OV, participants regarded SV as a better way to improve their confidence in public speaking, which is consistent with [5, 22]. Rickards-Schlichting et al. [22] asked participants to view manually edited self-videos that had been manually processed to eliminate verbal disfluencies and nervous behaviors, preserving only suitable and fluent expressions. The results showed that this self-modeling intervention decreased speech anxiety and increased the confidence of the participants. Our results indicated that generated videos may have the potential to achieve the impact of self-modeling intervention, demanding further validation in further investigations.

The results showed that the participants' posttest scores of speech pace improved significantly on the pretest after viewing CmV in comparison to viewing OV. The enhancement was also reflected in the semi-structured interview, where they indicated that CmV enhanced their speech delivery and increased their learning efficiency. In particular, participants were able to rapidly improve their public speaking performance by directly referencing the specifics of pace, tone, emotion, etc..

According to the results, we considered that the two kinds of videos, CmV and SV, contributed to improving public speaking performance within limited time. The exact method preferred depends on the participant's strategy. Some participants indicated that they preferred to learn from CmV. They believed that exemplary speakers in CmV provide a higher quality of modeling. They were more

inclined to learn from these role models since they were thought to be more persuasive in their public speaking. As Toubot et al. [27] pointed out that the lack of suitable role models is a contributing factor to speech anxiety. There were also several participants who referenced the uncanny valley effect of SV [19]. Thereby, they also preferred to view CmV. Nevertheless, some individuals preferred to learn by SV. They felt as if they had already completed the public speaking, which alleviated their nervousness and improved their self-confidence.

7 CONCLUSION

In order to assist individuals in upcoming public speaking, we explored methods to enhance public speaking performance within a limited time in this study. Three types of videos were provided to participants for learning and practicing: an original video (OV), a content-modified video (CmV) and a self-video (SV). Speech performances of participants were also evaluated before and after practice by qualified teachers. We found that viewing SV increased public speaking confidence significantly more than viewing OV. In the CmV condition, participants' speech pace improved significantly. Our data analysis indicated that CmV and SV were both useful for individuals. The method that the participants preferred to use was determined by their strategies.

8 LIMITATIONS & FUTURE DIRECTIONS

This study still has some limitations. In the semi-structured interviews, some participants indicated that the body language of speakers in synthesized videos was rigid and incongruent with the semantics; the video in SV lacked expression; due to factors such as hairstyle and clothing, there was still a gap between the appearance of the speaker in SV and that of the participant. Furthermore, many participants indicated that the script of public speaking is crucial for preparation and expressed a desire for the function of generating and revising scripts. Addressing these limits is a focus of our future work.

Another future direction is to broaden the range of public speaking types. Although we provided two types of speech in this pilot study, the scarcity of participants impeded a comprehensive analysis. Consequently, future works may focus on including more categories of public speaking, such as persuasive, informative, ceremonial and so on. The characteristics and practice methods of different types will be explored to effectively satisfy the requirements of different user groups.

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